

[DC] Navigating Impossible Spaces in Virtual Reality For Seamless Walking Experiences in Small Physical Spaces

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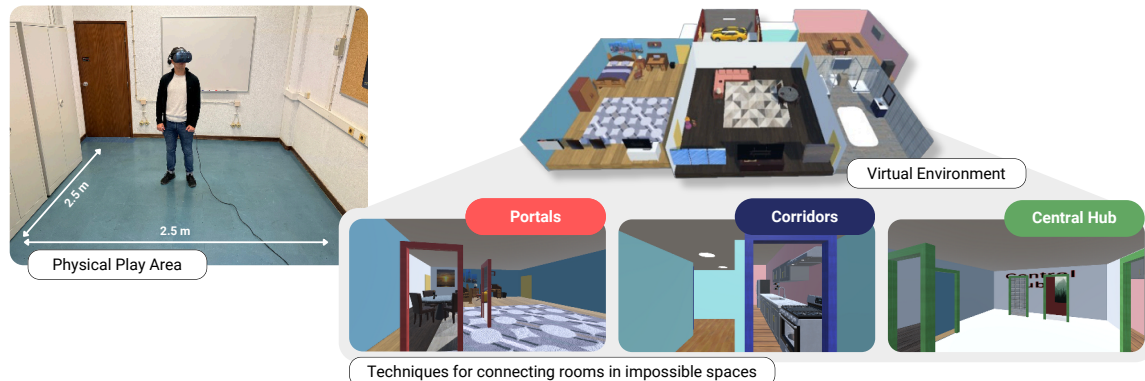


Figure 1: This research employs impossible spaces to enable walking within confined physical areas. Three techniques — portals, corridors, and central hubs — are studied to overlap multiple virtual spaces, maximizing the use of the physical play area. The case study focuses on virtual apartment tours.

ABSTRACT

In Virtual Reality (VR), navigating small physical spaces often relies on handheld controllers, such as teleportation or joystick movement, due to the limited space available for walking. However, walking-based techniques can enhance immersion by enabling more natural movement. This position paper presents research that employs the concept of “impossible spaces” to enable walking in small physical spaces. Three room-connection techniques – portals, corridors, and central hubs – are used to create impossible spaces by overlapping and adapting multiple virtual areas, maximizing the use of limited physical space. Our previous user studies show that all three techniques are viable for connecting rooms in VR within a play area of about 2.5 x 2.5 meters. Portals provide a flexible solution, as they can be placed in the middle of a room; corridors offer a seamless and natural transition between spaces; and central hubs simplify navigation in complex layouts by creating a central room that connects to all other rooms. The primary contribution of this work is to make walking in VR accessible for all users by demonstrating how these room-connection techniques can dynamically adapt virtual environments to fit small physical spaces, such as those commonly available to VR users at home.

Index Terms: Virtual reality, navigation, locomotion, impossible spaces.

1 INTRODUCTION

Effective VR experiences depend on user interaction with the virtual environment (VE), including how users navigate the virtual space and interact with virtual content. This research focuses on VR navigation, particularly in small physical play areas where walking and physical movement are often restricted.

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Typically, VR users have limited physical play areas, often around 2.4m x 2.2m [3], which restricts their ability to engage in natural walking. As a result, most VR applications rely on navigation techniques that use handheld controllers, such as teleportation and joystick movements, since these require less movement from the user. While effective, these techniques differ from the intuitive and natural movement experienced in the real world, which can limit immersion and usability [6]. Even advanced techniques such as redirected walking, face challenges in very small areas [4]. In response to this challenge, this research addresses the limitations of VR navigation in small physical spaces.

2 WALKING IN VR WITHIN SMALL PHYSICAL SPACES

The primary goal of this research is to develop guidelines and methods to enable comfortable walking in VR, envisioned to be integrated into a user-friendly tool that allows users to make the most of their VR experiences. This tool is designed to receive a VE the user wishes to explore and automatically adjust the environment to fit the available physical play area. By using techniques such as non-Euclidean and impossible spaces, which challenge the traditional linear way humans perceive the physical world, this tool creates VEs where multiple virtual rooms overlap within the same physical space [7], maximizing the use of the available physical area. The VEs adapt to the user’s surroundings, allowing for natural walking even in confined areas. This can be achieved with strategies such as portals, corridors, and stride adjustment (Figure 1). Portals are virtual doorways that connect two separate rooms, even if they are far apart in the real world [1, 9]. Corridors can change in shape and length, serving as transitional areas that guide the user smoothly from one room to another [8]. Adjusting the user’s stride — i.e., the distance covered with each step — helps users navigate larger virtual rooms within a limited physical space [2]. These strategies and additional adjustments will be applied based on the specific needs of each scenario, creating a personalized VE tailored to the application and the user’s characteristics. This selection process can be enhanced by Procedural Content Generation (PCG), a method for

automatically creating and customizing content using algorithms. In this context, PCG dynamically generates VEs that align with the user's physical space and preferences.

The practical applications of this VR tool ranges from virtual museum explorations to training simulations. Users can explore museums at their own pace and from the comfort of their homes, even if they have limited space, making tourist activities more accessible and enriching cultural experiences for a broader audience.

3 STUDIES AND FINDINGS

User Study 1: Our first user study compared traditional teleportation with a walking impossible space approach, which adjusts the virtual stride and utilizes portals to overlap multiple virtual areas within a small physical space (2.5×2.5 m). With 30 participants, the results revealed a strong preference (70%) for walking, as it provided a greater sense of presence and more natural navigation. The findings suggest that comfortable walking in VR environments up to 3.5 times larger than the physical space is achievable by tripling the stride. This shows that a virtual room, approximately 3.5 times larger than the physical area, can be accommodated while still offering a comfortable walking experience and maximizing the use of the play area.

User Study 2: In our second study, published in CHI 2024 [5], we adapted a 100 m² virtual apartment tour to fit a 6.25 m² physical play area, validating the idea of three room connection techniques – portals, corridors, and central hubs – through interviews with 10 HCI experts. This versatile system enables virtual tours, interactive e-commerce, and training simulations, even for emergency response. Usability challenges include adapting to various physical spaces, ensuring accessibility, and minimizing motion sickness. To enhance navigation and immersion, we collected feedback to include real-time customization, field-of-view modifications, safety alerts, and personalized soundscapes and lighting, creating a tailored and immersive experience across applications.

User Study 3: Our third user study evaluated the effectiveness of three room-connection techniques – portals, corridors, and central hubs. The study, conducted with 33 participants, evaluated whether these techniques contributed to a comfortable and immersive experience. Insights were gathered on presence, cybersickness, spatial awareness, orientation, and overall user experience. All three techniques reported positive results in these metrics, indicating that the choice of connection method does not significantly impact immersion. This finding suggests flexibility in selecting room-connection techniques based on user preferences or specific application requirements.

4 CONCLUSION AND FUTURE WORK

This paper presents ongoing research aimed at enabling walking in small physical spaces using impossible spaces. Our previous studies have shown encouraging results, revealing positive user feedback and comfortable walking experiences using our implementation of three room-connection techniques: portals, corridors, and central hubs.

Currently, user studies are being conducted to evaluate the adaptive component of the solution, specifically whether the tool generates a VE that prioritizes room-connection techniques that are appropriate to the application context and user preferences.

The next research objective focuses on navigating spaces that, even when divided into individual rooms, still exceed the user's available play area. To address these cases without relying on stride adjustments – an approach that, while effective for most users, can cause discomfort for those sensitive to motion sickness – we will explore alternative methods of subdividing and transitioning between sub-spaces. A key challenge lies in designing transitions that are seamless and intuitive. Potential solutions include techniques that encourage users to perform 90-180 degrees turns to re-center

within the play area, the use of temporary blackouts during repositioning, or visual cues such as arrows to guide the user back to the center of the space. This approach raises several questions that require further investigation. For instance, how might subdivided spaces affect the user's sense of presence? What design options could better preserve presence? Additionally, how might cybersickness levels vary when using these methods to navigate subdivided spaces?

Based on these challenges, the following research questions are proposed for further exploration:

- RQ1: What transitions can be used to enable seamless and intuitive navigation through sub-spaces?
- RQ2: How do presence and cybersickness impact navigating sub-spaces that match the play area size?

The primary contribution of this research is to provide guidelines and methods that allow walking in small physical spaces, enabling users to maximize their experience. VR applications can benefit significantly from these methods, offering users a more natural and engaging experience, even within the limited space typical of home settings.

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